

Supporting Information for “Susceptibility to Abrupt Thaw across Alaska’s Permafrost Landscapes”

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Introduction This file discusses the grid search used to select hyperparameters for the final XGBoost configuration (Text S1) and an analysis of out-of-fold skill and the dissimilarity index (Figure S1).

Text S1. Hyperparameters

We selected the XGBoost hyperparameters by grid search within the inner loop of the nested cross-validation described in Section 3.1 of the main text. The outer loop holds out spatial blocks to estimate skill, and within each outer training set we run a second five-fold spatial split to score every candidate configuration. We use 10-km equal-area blocks with no buffer. The search grid crosses five hyperparameters for 32 combinations in total: maximum tree depth (3, 5), minimum child weight (5, 20), L2 regularization (1, 10), learning rate (0.05, 0.1), and number of trees (200, 400). For each candidate, we measure the area under the precision-recall curve on the held-out inner folds and keep the highest scoring configuration.

This grid is deliberately narrow. Earlier exploration on previous versions of the labeled dataset searched a wider space that included deeper trees and row and column subsampling, but we set that aside in favor of spatially-blocked evaluation, given how costly nested cross-validation is.

The selection is not identical across folds, but consistently favors shallow and strongly regularized trees with a slow learning rate. The final configuration has a maximum depth of 3, minimum child weight of 5, L2 penalty of 1, a learning rate of 0.05, and 200 trees. For the logistic regression baseline model, we only tune the inverse regularization strength over 0.01, 0.1, and 1.0 in the same nested loop.

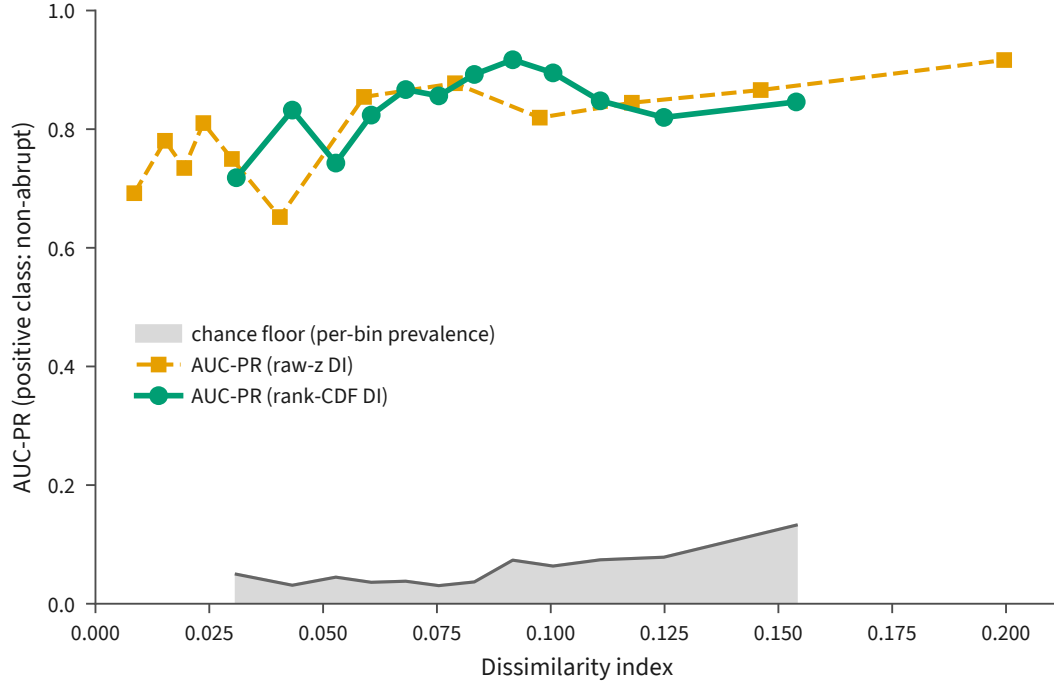


Figure S1. Model performance against dissimilarity index. Out-of-fold model skill does not degrade as dissimilarity in feature space rises. We plot pooled out-of-fold AUC-PR from the final model, computed under spatial cross-validation and binned into twelve equal-count bins by dissimilarity index, using non-abrupt as the positive class. Points sit at each bin’s median dissimilarity and the grey band is the per-bin chance floor. Across the sampled range, the rank-CDF (cumulative distribution function) AUC-PR stays between 0.72 and 0.92, well above the chance floor, and does not trend downward as dissimilarity grows. The green curve uses the operative rank-CDF dissimilarity index, while the orange curve repeats that analysis with the Meyer-Pebesma raw-z coordinate (Meyer & Pebesma, 2021). Dissimilarity does still track the magnitude of the model’s error, with a Spearman correlation of 0.49 between the rank-CDF index and the absolute out-of-fold residual.

References

- Meyer, H., & Pebesma, E. (2021). Predicting into unknown space? estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12(9), 1620–1633.